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REPUBLIC OF THE PHILIPPINES
BICOL UNIVERSITY
POLANGUI
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Polangui, Albay



IT 112: Web Systems and Technologies
BSIT- 2A 2nd Semester
S.Y. 2025-2026

Final Project – Phase 7: Testing, Documentation, and Final Deployment
Combat Coders: The End of Bugs Reign

Submitted by:

TEAM PLASMA

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Submitted to:

Dr. Guillermo V. Red, Jr, DIT

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Introduction

This week (Phase 7) focuses on the final phase of the Web Systems and Technologies project, which includes testing, refinement, documentation, and deployment of the developed web application. The main goal of this phase is to ensure that all features and system functions work properly and are ready for final presentation and evaluation.

During this phase, the group performs functional testing, improves the user interface and user experience, organizes the GitHub repository, and deploys the project to a live server using Render. Proper documentation is also prepared, including the project overview, implemented features, diagrams, testing results, and other important project details.

I. System Overview & Objective

Combat Coders is an educational game designed to teach the fundamentals of HTML, CSS, and JavaScript through an interactive fantasy-themed adventure. The game combines learning and gameplay by allowing players to battle enemies using coding knowledge as their weapon. Players progress through different adventure levels, complete coding quests, earn badges and rewards, and improve their understanding of web development concepts such as HTML tags, CSS properties, syntax, and stylesheet linking.

The main objective of the system is to promote learning through engagement and hands-on practice. Instead of using traditional tutorials, the game encourages players to answer coding-related questions to defeat enemies and advance through levels. Features such as hearts, hints, timers, leaderboards, and achievement badges motivate players to continue learning while improving their problem-solving and coding skills. Through its immersive gameplay and educational mechanics, Combat Coders aims to make web development learning more interactive, enjoyable, and effective.

II. Features implemented (CRUD, search, UI)

Game Modes

- **Adventure Mode** – Allows players to progress through multiple stages, including the Forest, Dungeon, and Volcano levels, by completing coding-based battles.
- **Quest Mode** – Provides 25 coding quests focused on HTML, CSS, and JavaScript concepts.
- **Learn Modules** – Offers instructional modules with short quizzes to reinforce learning and improve understanding of web development topics.
- **Character Selection** – Enables players to select from three playable characters: Witch, Knight, and Rogue.

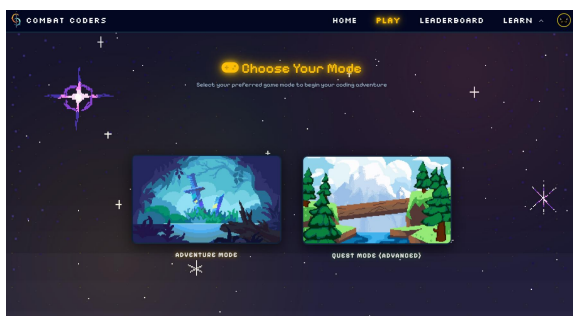


Image 1: Modes

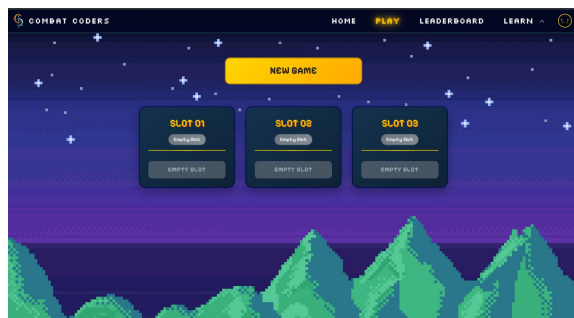


Image 2: Adventure Slots

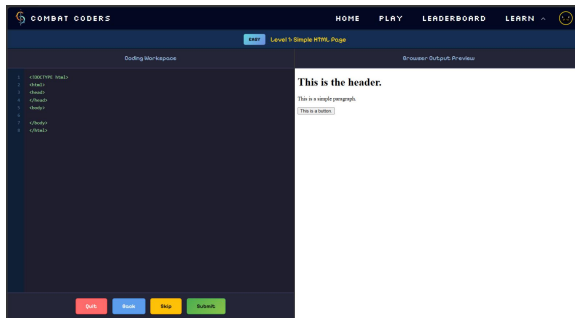


Image 3: Quests

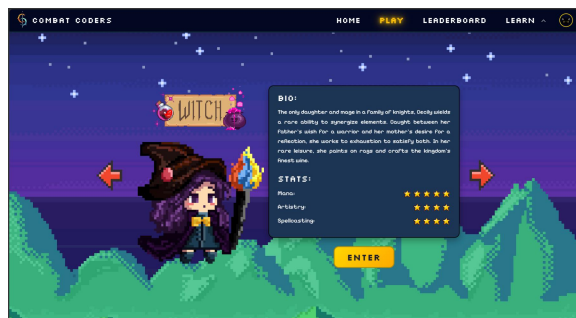


Image 4: Character Selection

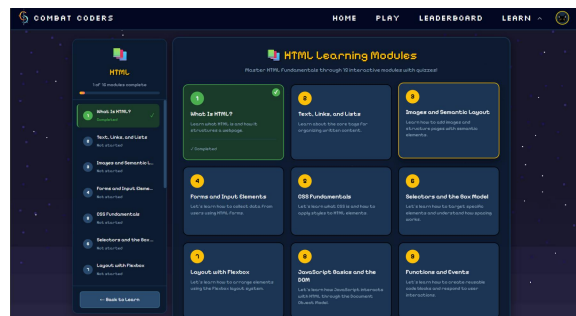


Image 5: Learning Modules

Player System

- **User Authentication** – Implements secure user registration and login functionality using JWT authentication tokens.
- **Profile Management** – Allows users to customize usernames and upload profile avatars with zoom and drag editing capabilities.
- **Avatar Presets** – Includes five predefined avatar options: Penguin, Raccoon, Kitty, Koala, and Cat.
- **Random Avatar Assignment** – Automatically assigns a random preset avatar to newly registered users.
- **Player Statistics** – Displays user information such as player level, experience points (XP), and registered email.

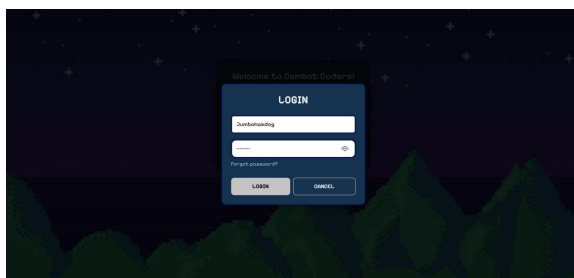


Image 6: User Authentication

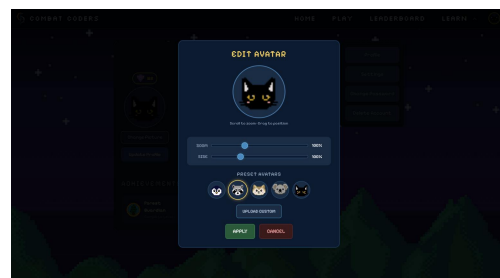


Image 7: Avatar Presets

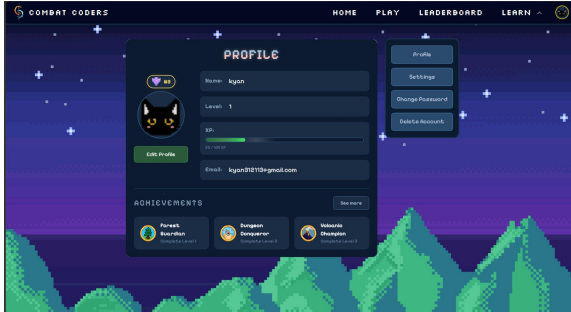


Image 8: Player Statistics

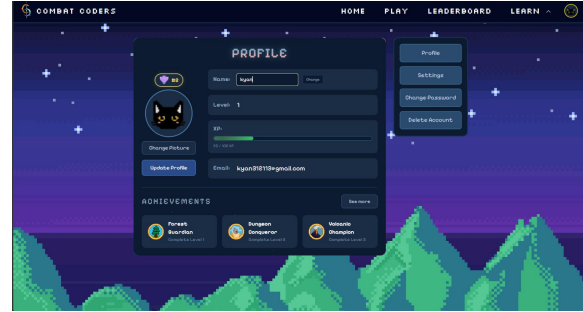


Image 9: Profile Management

Achievements and Progression

- **Quest Badges** – Awards one achievement badge for each completed quest, totaling 25 badges.
- **Level Medals** – Grants special medals for completing major stages, including Forest Guardian, Dungeon Conqueror, and Volcanic Champion.
- **XP System** – Provides experience points for completing quests and levels.
- **Level Progression** – Enables players to advance to higher levels based on accumulated XP.



Image 10: Level Medals



Image 11: XP System and Level Progression

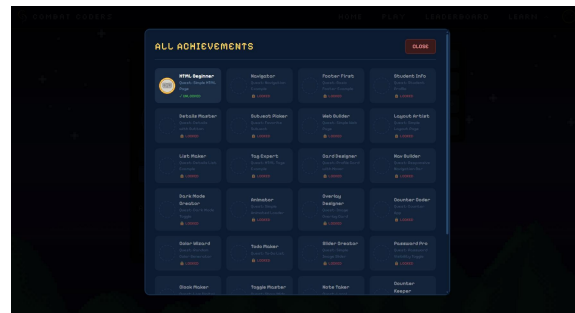


Image 12: Quests Medals

Leaderboard System

- **Adventure Leaderboard** – Ranks players according to their fastest completion time in Adventure Mode.
- **Quest Mode Leaderboard** – Displays rankings based on the highest level achieved in Quest Mode.
- **Panel Switcher** – Allows users to switch between leaderboard categories using navigation arrows.
- **Rank Display** – Assigns special rank icons to the top three players and standard rankings to other users.

- **Unranked Status** – Displays an “Unranked” label for players who have not yet completed the game.

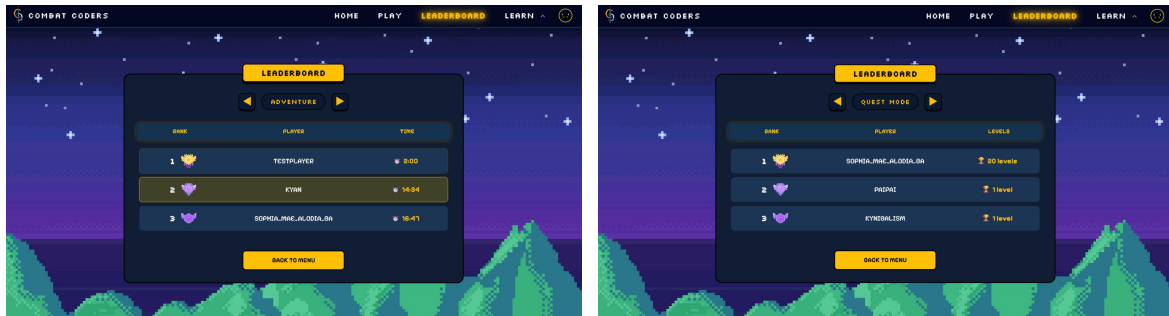


Image 13 and 14: Adventure and Quest Leaderboard with rankings.

UI/UX Features

- **Pixel Art Theme** – Utilizes pixel-style fonts such as Silkscreen and Pixelify Sans to maintain a retro gaming aesthetic.
- **Responsive Design** – Supports accessibility across mobile, tablet, and desktop devices.
- **Plasma Navigation Bar** – Provides consistent navigation throughout the application.
- **Learning Dropdown Menu** – Contains quick access to mechanics guides and HTML learning modules.
- **Settings Panel** – Includes options for sound control, password management, and account deletion.
- **Custom Scrollbars** – Implements styled scrollbars consistent with the game’s visual design.
- **Animations and Effects** – Incorporates leaderboard animations and particle effects to enhance user interaction and engagement.

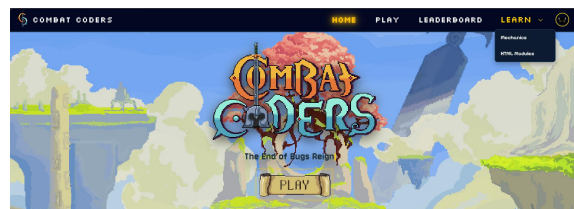


Image 15: Pixel Art Theme with Dropdown Menu

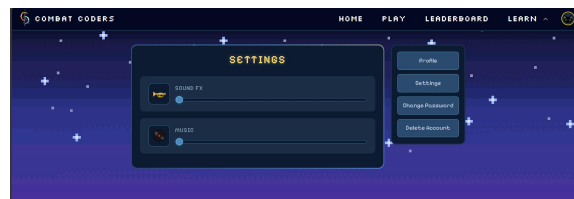


Image 16: Settings

CRUD Operations

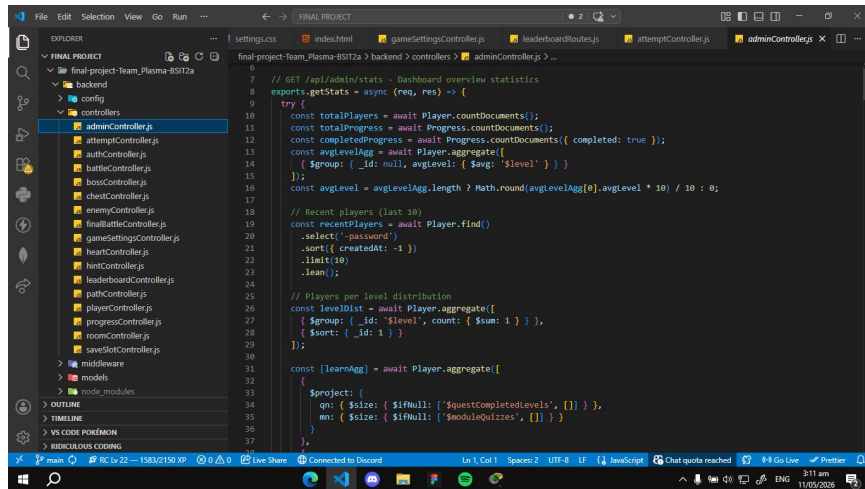


Image 17

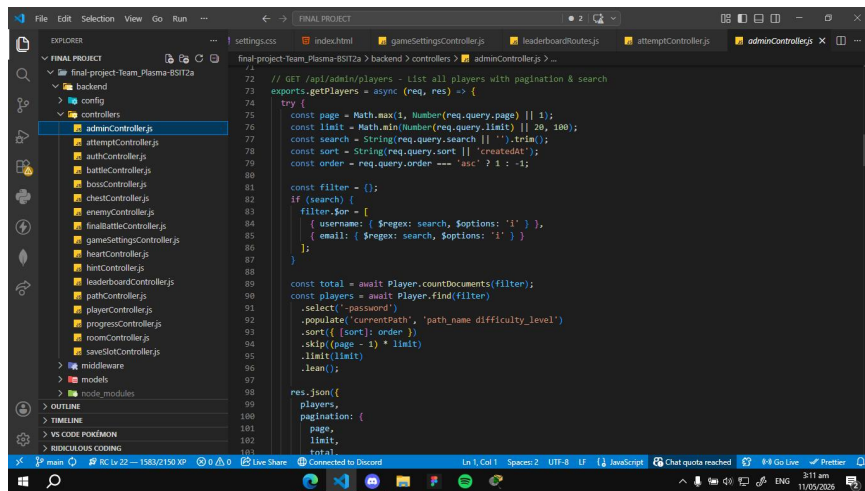


Image 18

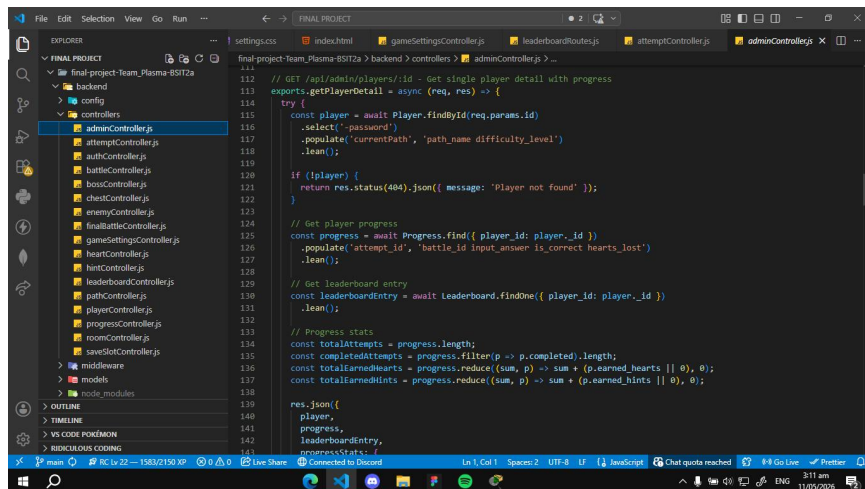


Image 19

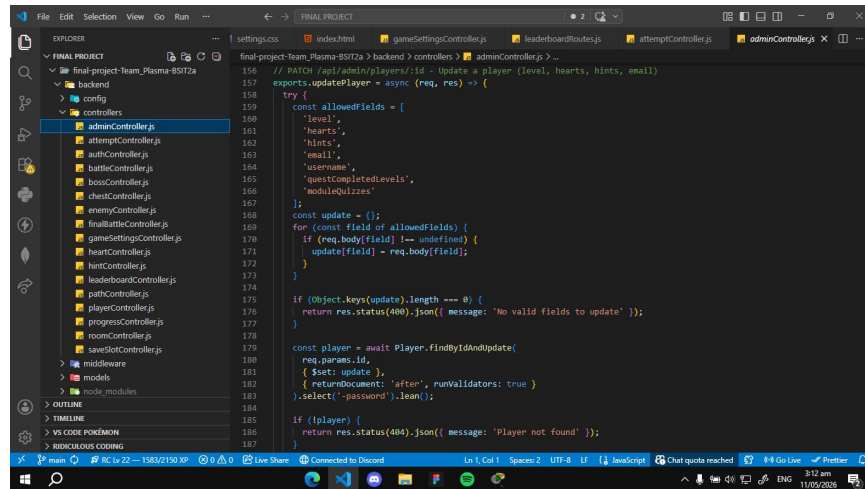


Image 20

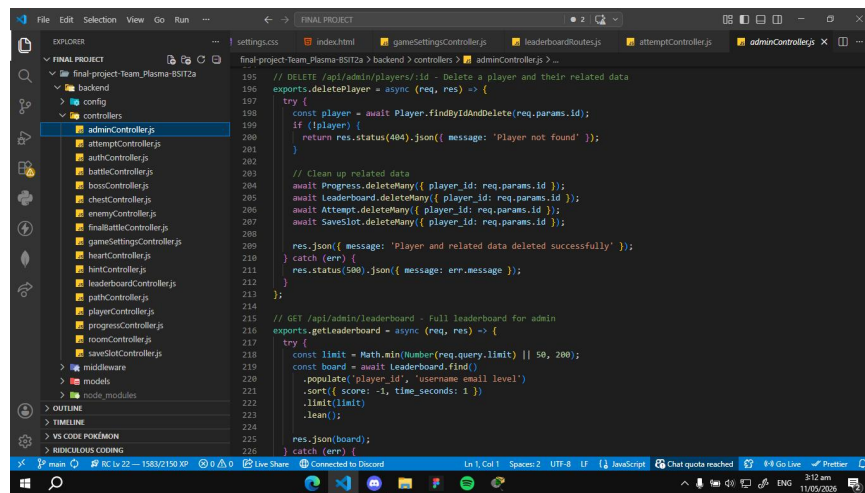


Image 21

III. Architecture Diagram (Frontend–Backend–Database flow)

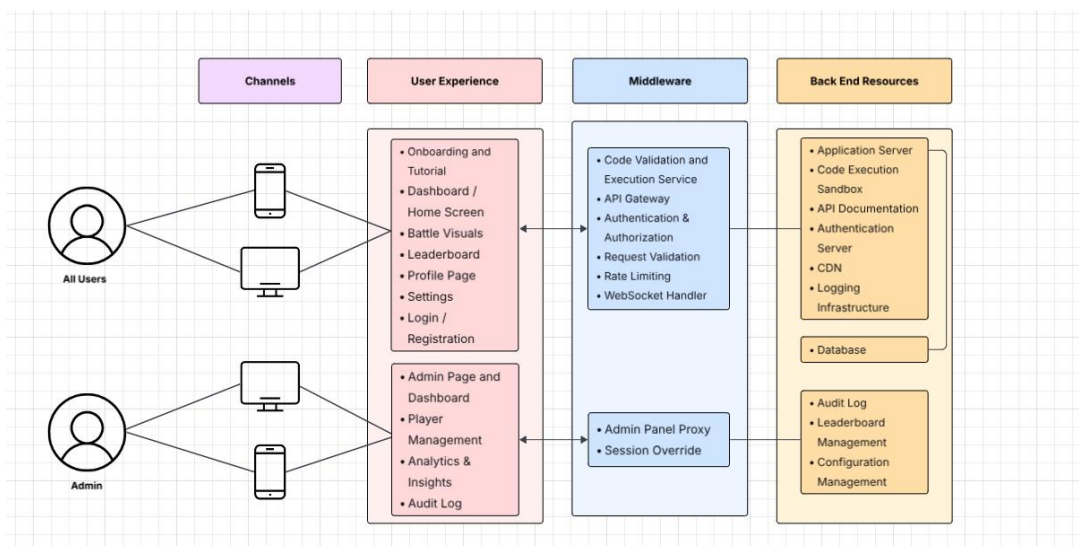


Image 22: Architecture Diagram

IV. Entity Relationship Diagram (ERD)

Combat Coders - ENTITY-RELATIONSHIP DIAGRAM

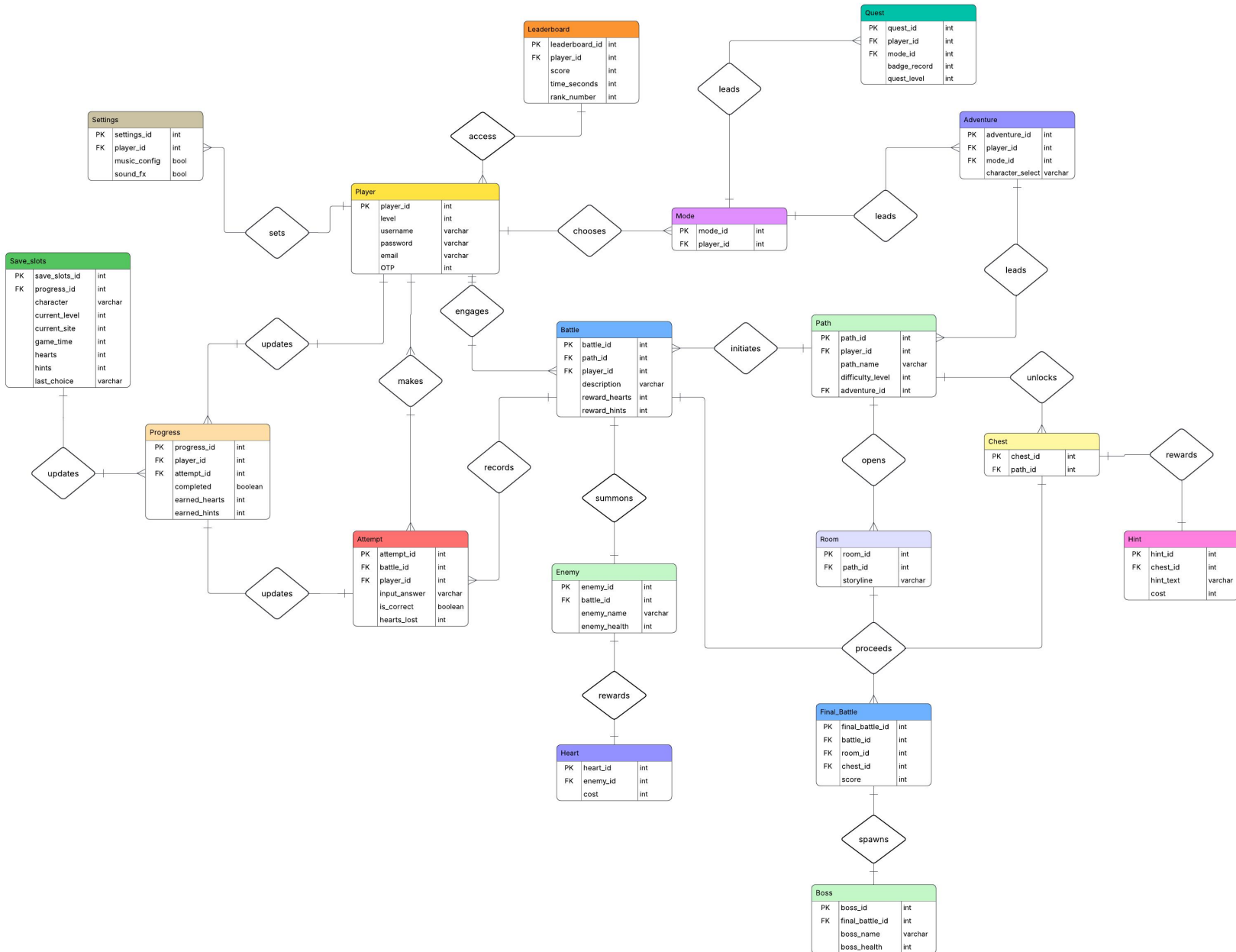


Image 23: ERD Diagram

V. GitHub Link + Commit Logs Summary

GitHub Link: https://github.com/hafsaharbotuzon24-lgtm/final-project-Team_Plasma-BSIT2a

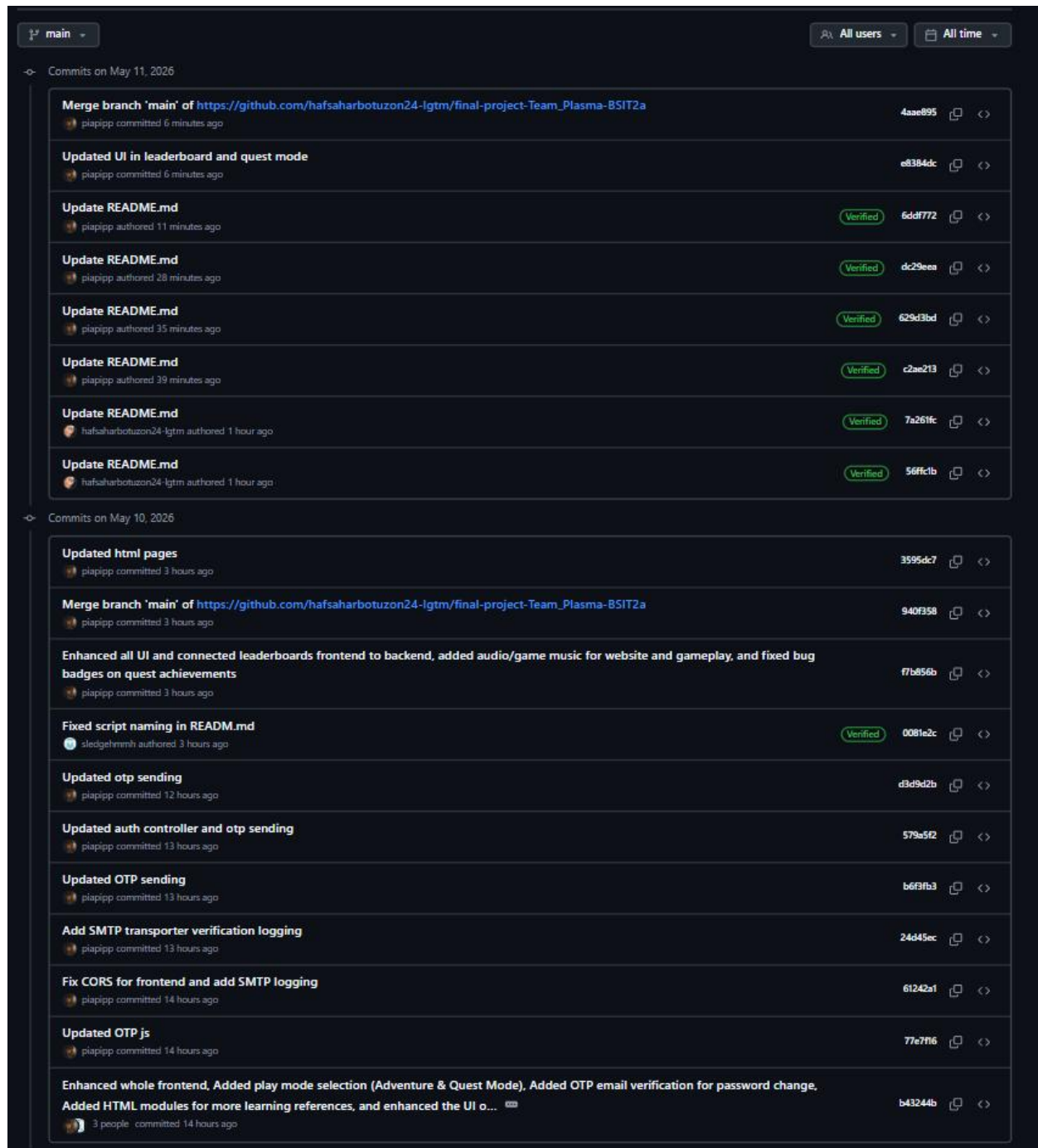


Image 24: GitHub Commits

VI. Render Link / Screenshots

Render Link: <https://combat-coders-team-plasma-frontend.onrender.com/>

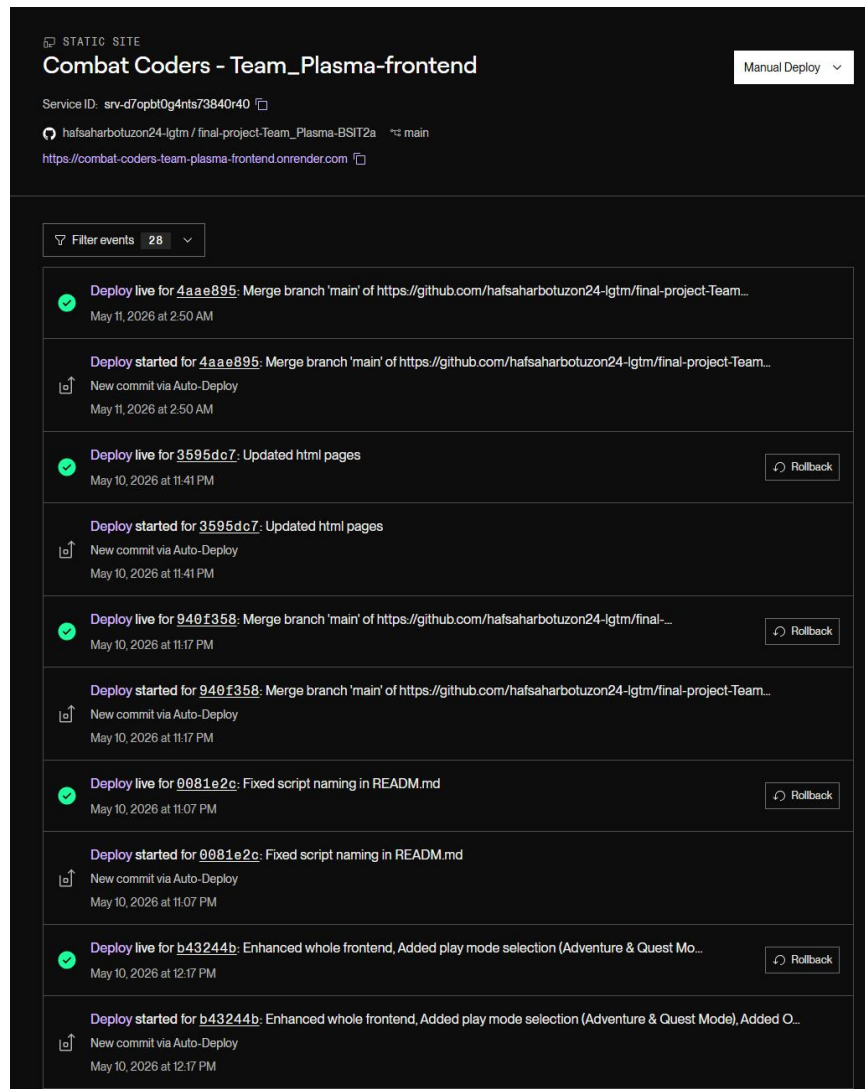


Image 25

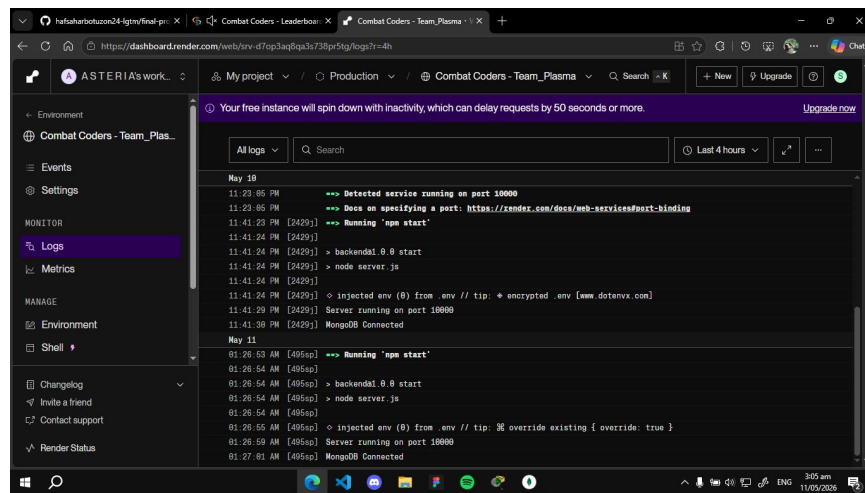


Image 26

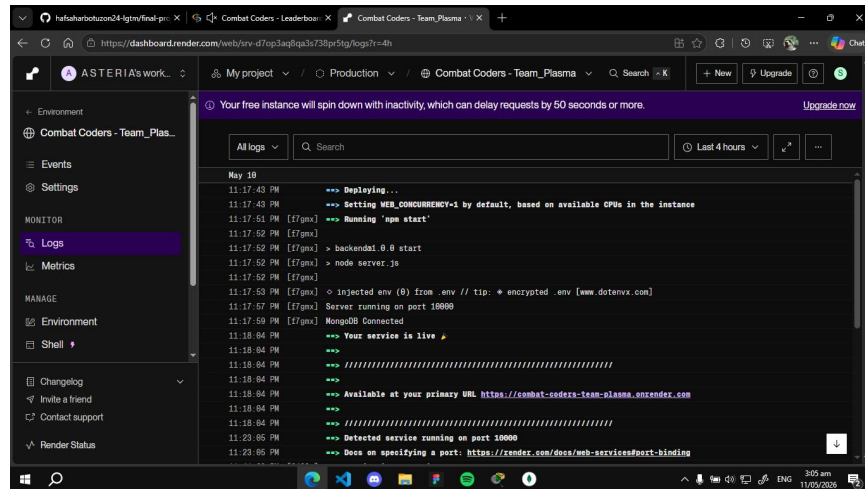


Image 27

VII. Summary of Division of Tasks

Sophia Mae Alodia J. Balmaceda: *Backend Developer and Project Manager*

- **Full-Stack Integration:** Connected frontend to backend across multiple pages; properly linked deployed frontend and backend environments using Render and environment files.
- **CRUD & Admin:** Implemented CRUD operations for player data and the admin dashboard.
- **Email Service:** Integrated OTP email sending through Brevo.
- **Game Mode:** Added play mode selection with a new Quest mode, where players recreate a given website preview.
- **Debugging:** Conducted bug testing through console logs.

Dwayne Lester O. Diaz: *Frontend Developer*

- **Audio Integration:** Added background music and sound effects across all pages; built an audio manager to loop page-specific tracks and click sounds on interactive elements.
- **UI Improvements:** Enhanced the leaderboard's appearance and fixed avatar and achievement display issues.
- **Responsive Design:** Adjusted the layout for better mobile responsiveness.

Kian Carlo O. Matutina: *Documentation Officer*

- **QA & Testing:** Tested the game for errors during gameplay and made minor code adjustments to refine questions.
- **Battle Content:** Finalized questions, answers, and hints for each boss.
- **Player Customization:** Created player avatars, icons for in-game selection and medals.
- **Documentation:** Served as documentation officer; took meeting notes, captured screenshots for project records, and led the preparation and finalization of the laboratory report.

Hafsah A. Tuzon: *GitHub Manager*

- **Visual Documentation:** Captured screenshots for the lab report and README file.
- **Design Enhancement:** Redrawn original designs to improve visual quality of project materials.
- **Report & Presentation:** Assisted in creating the lab report and final PowerPoint presentation.

- **Quality Assurance:** Revised all project resources for accuracy and consistency.

Joshua Cedric B. Velitario: : *Database Manager*

- **QA & Testing:** Tested the UI and gameplay for bugs; verified MongoDB account creation and deletion.
- **Battle Content:** Wrote questions, answers, and hints for the battle system.
- **Narrative Design:** Created in-game dialogue, including modals, scenery text, and character interactions.
- **Design & Drawing:** Contributed to the visual design and drawing of game elements.
- **Documentation:** Assisted with the laboratory report and took the necessary screenshots.

VIII. Bug testing results / final test notes

The system underwent final functional and usability testing to ensure that all features operated correctly in both local and deployed environments. Testing was conducted on major functionalities such as user authentication, player profile management, CRUD operations, game progression, leaderboards, quest systems, and responsive user interface components. Both valid and invalid inputs were tested to verify proper error handling and data validation.

Results showed that the application functioned successfully with no critical errors affecting system performance or usability. All API routes and database operations were confirmed to work correctly, including creating, retrieving, updating, and deleting records. The user interface displayed properly across different screen sizes, and navigation, animations, forms, and interactive components behaved as expected. Minor visual and alignment issues identified during testing were resolved prior to final deployment. Overall, the system was verified to be stable, functional, and ready for final presentation and deployment.

IX. Short Reflections From Each Member

Sophia Mae Alodia J. Balmaceda: *Backend Developer and Project Manager*

Helped with multiple frontend pages while simultaneously connecting everything from frontend to backend. From CRUD implementation for player data and admin dashboard, to bug testing in console logs, and connecting OTP Email sending through Brevo. I also added a play mode selection with a new game: Quest mode, where you have to recreate the given website preview. I also handled the proper use of environment files to our deployed render for backend and properly connected it to the deployed frontend.

Dwayne Lester O. Diaz: *Frontend Developer*

During this phase, I focused on enhancing the UI of Combat Coders by adding background music and sound effects across all pages. I built an audio manager that plays different music tracks depending on the page and loops them continuously, along with click sounds on interactive elements. I also improved the leaderboard's appearance, fixed the avatar and achievement display issues, and made the layout more responsive on mobile devices.

Joshua Cedric B. Velitario: *Database Manager*

During this project, I contributed to gameplay logic, narrative design, visual design, and report preparation. I tested the UI and gameplay for bugs, verified MongoDB account creation and deletion, and wrote questions, answers, and hints for the battle system. I also created in-game dialogue, including modals, scenery text, and character interactions. Additionally, I took part in designing and drawing game elements. Finally, I assisted with the laboratory report and took the

necessary screenshots. These efforts helped make the game feel polished, functional, and engaging.

Kian Carlo O. Matutina: *Documentation Officer*

During this phase, I tested the game for errors during gameplay and made minor code adjustments to refine the questions. I also finalized the questions, answers, and hints for each boss, as well as the easy part of the newly added quests mode, and created the medals for some of the levels of the quests mode. As the documentation officer, I took meeting notes, captured screenshots for project records, and led the preparation and finalization of the laboratory report.

Hafsah A. Tuzon: *GitHub Manager*

I created and checked the majority of the project designs, specifically the backgrounds, icons, sprites, and visual elements for each game scenario, using Piskel for several of the backgrounds. I helped improve the overall visual design of the game and assisted in writing and completing the laboratory report. I also organized and finalized the screenshots provided by the project leader for the final document. Overall, I played a key role in both the visual and technical parts of the report.